Poisson–Gaussian transport: closed graphs, coherent recovery and nonlinear limits

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Abstract

The Charlier–Hermite correspondence identifies Poisson and Gaussian Hilbert spaces and their complete annihilation and creation graphs. We reconstruct the basis, maximal domains and graph-core arguments, then transport actual Poisson changes of intensity into a strongly continuous Gaussian displacement group with an explicit weighted-shift formula on the whole L^2 space. The correspondence preserves distances between square-root likelihoods and their derivative energies, but it does not preserve positivity, ordinary multiplication, or entropy. Coherent vectors give quantitative entropy distortion and exclude a bounded realization of any nonzero displacement on L^4. A compact creation inverse and a sharp shifted Poincaré estimate hold on the original maximal domains. A compensated-count example separates the small- and large-intensity $L^{4/3}$ scales. Positive pointwise jump curvature coexists with this entropy failure, and the missing diffusion chain rule is exhibited at one atom. The failure of a quadratic Poisson logarithmic Sobolev inequality already occurs on finite-support functions; we prove this by controlling the cutoff boundary in the original difference graph. These are classical mechanisms, consolidated as a companion to frozen verified constructions. No new Fock-space correspondence or logarithmic Sobolev failure is claimed.

1 What is transported

Fix $\lambda > 0$. Write

$$p_\lambda(k) = e^{-\lambda}\frac{\lambda^k}{k!}, \quad k \in \mathbb{N}, \qquad d\gamma(x) = \frac{e^{-x^2/2}}{\sqrt{2\pi}}\,dx,$$

and use real Hilbert spaces $H_\lambda = L^2(p_\lambda)$ and $H_G = L^2(\gamma)$. Since every Poisson atom has positive mass, an element of H_λ has a uniquely determined value at every integer. Gaussian functions are equivalence classes modulo γ-null sets. The scalar normalization of the Poisson annihilation operator is

$$A_\lambda f(k) = \sqrt{\lambda}\,[f(k+1) - f(k)]. \tag{1}$$

Its domain consists of precisely those $f \in H_\lambda$ for which the displayed sequence belongs to H_λ. We will construct a unitary $U_\lambda : H_\lambda \to H_G$ and prove, on full domains,

$$U_\lambda A_\lambda = DU_\lambda, \qquad U_\lambda A_\lambda^* = \delta U_\lambda, \quad D = \partial_x, \quad \delta = x - \partial_x.$$

This is an identification of Hilbert spaces with closed operators. It supplies no identification of their ordinary products or laws.

The correspondence is classical. Vershik and Tsilevich [1, Sections 1.3 and 3.1] give the Poisson–Gaussian isometry, including its Hermite–Charlier specialization, and explicitly discuss

its failure to be positive or multiplicative. The quadratic Poisson entropy obstruction is also classical; the introduction and Theorem 1.1 of Aryan–López-Rivera–Shenfeld [2] place it beside Wu's modified inequality. Our contribution is a complete exposition tied to the original verified graphs, gauges and counterexamples, with these antecedents distinguished from claims of priority.

2 Complete bases and the unitary correspondence

Define the Charlier and probabilists' Hermite polynomials by

$$e^{-\lambda t}(1+t)^k = \sum_{n\geq 0} C_n^\lambda(k)\frac{t^n}{n!}, \qquad e^{tx-t^2/2} = \sum_{n\geq 0} \mathrm{He}_n(x)\frac{t^n}{n!}. \tag{2}$$

Thus $C_0 = 1$, $C_1 = k - \lambda$, and $C_2 = (k-\lambda)^2 - (k-\lambda) - \lambda$. Put

$$q_n^\lambda = \frac{C_n^\lambda}{\sqrt{n!\lambda^n}}, \qquad h_n = \frac{\mathrm{He}_n}{\sqrt{n!}}.$$

Lemma 2.1. *The sequences (q_n^λ) and (h_n) are complete orthonormal bases of H_λ and H_G, respectively. There is exactly one unitary U_λ sending every q_n^λ to h_n.*

Proof. For N of law p_λ, the expectation of the product of the first generating functions at s, t is

$$e^{-\lambda(s+t)}\mathbb{E}[(1+s)^N(1+t)^N] = e^{\lambda st}.$$

The analogous Gaussian expectation is e^{st}. Differentiating at zero gives orthogonality with squared norms $n!\lambda^n$ and $n!$. All differentiations are justified locally by exponential integrability. Both polynomial sequences are monic and have degree n, so their algebraic spans are the polynomial spaces.

To prove Poisson completeness, suppose $f \in H_\lambda$ is orthogonal to all polynomials. The series

$$F(z) = \sum_{k\geq 0} f(k)p_\lambda(k)z^k$$

is entire: on $|z| \leq R$ its absolute sum is bounded by $\|f\|_2 \exp\{\lambda(R^2-1)/2\}$, and its tails tend to zero uniformly by Cauchy–Schwarz. Every derivative at $z = 1$ vanishes, because falling factorials are polynomials. Thus $F = 0$, and every coefficient $f(k)p_\lambda(k)$ is zero.

For Gaussian completeness, if f is orthogonal to every polynomial, the entire function $L(z) = \int f(x)e^{zx}\,d\gamma(x)$ has all derivatives at zero equal to zero. Entirety and differentiation follow by Cauchy–Schwarz and Gaussian exponential moments, uniformly on compact sets. Hence $L = 0$. The finite signed measure $f\,d\gamma$ has zero Fourier transform and is zero, so $f = 0$. For completeness, Fourier uniqueness here follows by convolution with a Gaussian approximate identity: Fourier inversion gives a zero convolution at every positive variance, and the approximate identity converges weakly to the original finite measure. Finally extend the isometry between the two orthonormal bases by Hilbert completion; density gives both surjectivity and uniqueness. □

For $f_n = \langle q_n^\lambda, f\rangle$, the construction gives

$$U_\lambda f = \sum_{n\geq 0} f_n h_n, \qquad \|f\|_2^2 = \sum_{n\geq 0} |f_n|^2. \tag{3}$$

Every series in this paper that represents an arbitrary L^2 function is understood in its Hilbert norm, rather than pointwise.

Proposition 2.2. *The unitary U_λ is neither positivity preserving nor multiplicative for ordinary products, even on polynomials.*

Proof. It sends the nonnegative function k to $\lambda + \sqrt{\lambda}x$, which is negative on a set of positive Gaussian measure. For $r(k) = (k-\lambda)/\sqrt{\lambda}$, we have $U_\lambda r = x$ whereas the identity for C_2 gives

$$U_\lambda(r^2) = x^2 + \frac{x}{\sqrt{\lambda}} \neq (U_\lambda r)^2.$$

All functions and products used here belong to L^2. □

3 The maximal graphs, including their endpoints

Define the maximal creation expression by

$$A_\lambda^* g(k) = \frac{k}{\sqrt{\lambda}} g(k-1) - \sqrt{\lambda} g(k), \tag{4}$$

where the first term is zero at $k = 0$. Its domain is the set of $g \in H_\lambda$ for which this entire expression belongs to H_λ; separate square integrability of its two terms is not required. Both maximal graphs are closed. Indeed, convergence in H_λ implies convergence at every fixed atom, since $|f(k)| \le p_\lambda(k)^{-1/2}\|f\|_2$. One may therefore pass to the pointwise expressions in any convergent graph sequence.

Theorem 3.1. *For $f, g \in H_\lambda$ with coefficients $(f_n), (g_n)$,*

$$A_\lambda f = g \iff g_n = \sqrt{n+1}\, f_{n+1} \quad (n \ge 0), \tag{5}$$

$$A_\lambda^* g = f \iff f_0 = 0, \quad f_{n+1} = \sqrt{n+1}\, g_n \quad (n \ge 0). \tag{6}$$

Consequently

$$\mathrm{Dom}\, A_\lambda = \left\{ f : \sum_{n\ge 0} n|f_n|^2 < \infty \right\}, \qquad \|A_\lambda f\|_2^2 = \sum_{n\ge 0} n|f_n|^2, \tag{7}$$

$$\mathrm{Dom}\, A_\lambda^* = \left\{ g : \sum_{n\ge 0} (n+1)|g_n|^2 < \infty \right\}, \qquad \|A_\lambda^* g\|_2^2 = \sum_{n\ge 0} (n+1)|g_n|^2. \tag{8}$$

These operators are Hilbert adjoints of one another, and finite Charlier expansions are graph cores for both.

Proof. The generating function gives $\Delta C_{n+1}^\lambda = (n+1)C_n^\lambda$ and hence $A_\lambda q_{n+1}^\lambda = \sqrt{n+1} q_n^\lambda$. Summation by parts uses exactly $(k+1)p_\lambda(k+1) = \lambda p_\lambda(k)$, including the zero endpoint in (4). Against a polynomial test it yields

$$\langle A_\lambda f, v\rangle = \langle f, A_\lambda^* v\rangle, \qquad \langle A_\lambda^* g, v\rangle = \langle g, A_\lambda v\rangle$$

for the respective maximal-domain inputs. The reindexed sums are absolutely convergent: each reduces to a product of an L^2 input and a polynomial in $L^2(p_\lambda)$. Thus no boundary term at infinity is discarded. Applying the first identity to q_n^λ and the second to q_{n+1}^λ gives the coefficient rules. The test 1 also gives $f_0 = 0$ in the creation rule.

Conversely, assume the coefficient rule (5). Truncate f at indices $0, \dots, M$ and g at $0, \dots, M-1$. The finite ladder identities give the annihilation graph at every truncation; both components converge by completeness. Closedness proves the full graph relation. For (6), truncate g at $0, \dots, M-1$ and f at $0, \dots, M$, and use the same argument. This also proves the stated domain formulas and graph-core property.

In coefficient coordinates the annihilation operator is the weighted backward shift $(f_n) \mapsto (\sqrt{n+1} f_{n+1})$. Its Hilbert adjoint is precisely the weighted forward shift in (6): finite sequences determine the adjoint coordinates, and their square summability is necessary and sufficient for a bounded functional on ℓ^2. This proves actual adjointness on the full domains, rather than just the polynomial integration-by-parts identity. □

Lemma 3.2. *Finite-support sequences are also a common graph core for A_λ and A_λ^*.*

Proof. Write $f^{[M]}(k) = f(k)\mathbf{1}_{\{k<M\}}$. For $f \in \operatorname{Dom} A_\lambda$, the shifted sequence $f(k+1) = f(k) + A_\lambda f(k)/\sqrt{\lambda}$ is square integrable. Reindexing shows $\sum_k k p_\lambda(k) f(k)^2 < \infty$. Thus

$$\|A_\lambda f^{[M]} - A_\lambda f\|_2^2 = M p_\lambda(M) f(M)^2 + \sum_{k \geq M} p_\lambda(k) |A_\lambda f(k)|^2 \longrightarrow 0.$$

For $g \in \operatorname{Dom} A_\lambda^*$, direct evaluation at the single boundary atom gives instead

$$\|A_\lambda^* g^{[M]} - A_\lambda^* g\|_2^2 = \lambda p_\lambda(M) g(M)^2 + \sum_{k > M} p_\lambda(k) |A_\lambda^* g(k)|^2 \longrightarrow 0.$$

Both scalar cutoffs converge in H_λ, and their operator expressions have finite support. These identities prove graph density in both domains, including the boundary contributions. □

Theorem 3.3. *Let D be the maximal weak derivative in $L^2(\gamma)$ and let $\delta g = xg - g'$ in the distributional sense, with maximal L^2 domain. Then*

$$(f, g) \in \operatorname{graph} A_\lambda \iff (U_\lambda f, U_\lambda g) \in \operatorname{graph} D,$$

and the same equivalence holds with A_λ^* and δ.

Proof. The Gaussian generating function gives $D h_{n+1} = \sqrt{n+1} h_n$ and $\delta h_n = \sqrt{n+1} h_{n+1}$. The weak graphs are closed: Gaussian L^2 convergence implies local Lebesgue L^1 convergence, so the distributional equations pass to limits. For a maximal weak graph, Gaussian integration by parts against $h_n \chi_R$, where χ_R is a smooth cutoff equal to one on $[-R, R]$, gives the same coefficient rules as (5)–(6) as $R \to \infty$. The polynomial terms converge in $L^2(\gamma)$; the cutoff-derivative term tends to zero by Cauchy–Schwarz and Gaussian polynomial tails. This argument applies to $xg - g'$ as one distribution, without requiring xg and g' separately to belong to L^2. Conversely, finite Hermite truncations and closedness give both maximal weak graphs from their coefficient rules. Their domains therefore coincide exactly with the two weighted sequence domains in Theorem 3.1. Equation (3) now proves both equivalences. □

In the frozen formalization the original difference is Δ, with field space $L^2(\lambda p_\lambda)$. Multiplication of a field by $\sqrt{\lambda}$ identifies that space with H_λ and produces (1). This normalization is essential for the graph transport.

4 Likelihood roots and the physical change of intensity

For $\theta \in \mathbb{R}$ put $\lambda_\theta = \lambda e^\theta$ and

$$r_{\lambda,\theta}(k) = \exp\left\{\frac{\theta k}{2} - \frac{\lambda(e^\theta - 1)}{2}\right\}, \quad a = \sqrt{\lambda}(e^{\theta/2} - 1), \quad g_a(x) = e^{ax - a^2}. \tag{9}$$

The square of $r_{\lambda,\theta}$ is the actual likelihood ratio $p_{\lambda_\theta}/p_\lambda$; the square of g_a is the likelihood ratio of $N(2a, 1)$ to $N(0, 1)$. Both roots have norm one.

Proposition 4.1. *One has $U_\lambda r_{\lambda,\theta} = g_a$. For any two tilt parameters,*

$$\begin{aligned} \langle r_{\lambda,\theta}, r_{\lambda,\eta}\rangle &= \exp\left\{-\tfrac{\lambda}{2}(e^{\theta/2} - e^{\eta/2})^2\right\}, \\ \|r_{\lambda,\theta} - r_{\lambda,\eta}\|_2^2 &= 2\left[1 - \exp\left\{-\tfrac{\lambda}{2}(e^{\theta/2} - e^{\eta/2})^2\right\}\right]. \end{aligned} \tag{10}$$

Thus the correspondence preserves the exact Hellinger distances within this family, under the square-root-intensity coordinate.

Proof. Write $s = e^{\theta/2}$, so $a = \sqrt{\lambda}(s-1)$. Poisson summation of the first generating function gives

$$\mathbb{E}\big[r_{\lambda,\theta}(N)e^{-\lambda t}(1+t)^N\big] = e^{-a^2/2}e^{\sqrt{\lambda}at}.$$

Therefore the nth normalized Charlier coefficient of the root is $e^{-a^2/2}a^n/\sqrt{n!}$. These are precisely the normalized Hermite coefficients of g_a, by the second generating function. Coefficient equality in the complete bases proves the identity. Directly summing the product of the two Poisson roots gives (10). The norm formula follows because both roots have norm one. Alternatively, the Gaussian identity $\langle g_a, g_b\rangle = e^{-(a-b)^2/2}$ gives the same calculation after transport. □

The half-density change of measure is the actual unitary

$$V_{\lambda,\theta} : H_{\lambda_\theta} \longrightarrow H_\lambda, \qquad V_{\lambda,\theta} f = r_{\lambda,\theta} f. \tag{11}$$

Indeed, $\|Vf\|^2_{H_\lambda} = \|f\|^2_{H_{\lambda_\theta}}$, and division by the strictly positive root is the inverse.

Proposition 4.2. *On the complete maximal graphs,*

$$A_\lambda V f = V A_{\lambda_\theta} f + aVf, \qquad \operatorname{Dom} A_\lambda = V \operatorname{Dom} A_{\lambda_\theta}, \tag{12}$$

$$A^*_\lambda V f = V A^*_{\lambda_\theta} f + aVf, \qquad \operatorname{Dom} A^*_\lambda = V \operatorname{Dom} A^*_{\lambda_\theta}. \tag{13}$$

Proof. Use $r(k+1) = e^{\theta/2} r(k)$ in (1) and $r(k-1) = e^{-\theta/2} r(k)$ in (4). The two pointwise identities follow by expansion, including $k = 0$. Since V is unitary, the extra term aVf is always in H_λ. Consequently either side has an L^2 operator output exactly when the other does. This proves both domain equalities and both graph identities. □

5 A group on the entire Gaussian space

Define

$$W_a = U_\lambda V_{\lambda,\theta} U^{-1}_{\lambda_\theta}, \qquad a = \sqrt{\lambda}(e^{\theta/2} - 1). \tag{14}$$

Every $a \in \mathbb{R}$ is attainable: choose $b > |a|$, take $\lambda = b^2$ and $\theta = 2\log((b+a)/b)$. The following result proves that (14) is independent of this choice.

Theorem 5.1. *For every $f \in H_G$, the whole-space formula is*

$$(W_a f)(x) = e^{ax-a^2} f(x-2a) \qquad \text{for almost every } x. \tag{15}$$

These operators form a strongly continuous unitary group: $W_a W_b = W_{a+b}$, $W_0 = I$. They satisfy, on exact domains,

$$DW_a f = W_a Df + aW_a f, \qquad \delta W_a f = W_a \delta f + aW_a f. \tag{16}$$

Proof. First the right side of (15) defines a unitary T_a. The change of variables $y = x - 2a$ gives

$$e^{2ax-2a^2} e^{-x^2/2} = e^{-(x-2a)^2/2}, \qquad \|T_a f\|_2^2 = \|f\|_2^2.$$

All Gaussian null sets are Lebesgue null sets, and translations preserve them; hence the formula is well defined on equivalence classes. The same formula gives $T_a^{-1} = T_{-a}$.

We identify T_a with the unitary in (14) without assuming a polynomial rule extends to a nonlinear expression. For $s = \sqrt{\lambda_\theta}$ and any $b > -s$, choose η so that $b = s(e^{\eta/2} - 1)$. Proposition 4.1 and the exact cocycle

$$r_{\lambda,\theta}\, r_{\lambda_\theta,\eta} = r_{\lambda,\theta+\eta}$$

give $W_a g_b = g_{a+b}$. Direct substitution gives the same identity for T_a. The vectors g_b with b in any nonempty open interval have dense linear span in H_G: orthogonality to them makes the entire function $\int f(x) e^{zx}\, d\gamma(x)$ vanish on that interval; the identity theorem and Fourier uniqueness then give $f = 0$. Thus the bounded operators W_a, T_a agree on a dense set and everywhere.

Multiplying the two explicit weights in $T_a T_b$ gives $e^{(a+b)x-(a+b)^2}$ and translation by $2(a+b)$, proving the group law and $W_0 = I$. Also

$$\|W_a g_b - g_b\|_2^2 = 2(1 - e^{-a^2/2}) \longrightarrow 0.$$

Finite coherent linear combinations are dense, and the W_a are isometries, so this proves strong continuity at zero for every f. The group law proves it at every parameter. Finally (16) follows from Theorem 3.3 and Proposition 4.2, including domain equivalence. It can also be checked by distributional differentiation of (15); the inverse shift then gives the converse domain inclusion. □

Strong continuity does not mean continuity in operator norm. For $a > 0$, let f_a be the indicator of $(0, a)$ divided by its Gaussian L^2 norm. Its displaced image is supported on $(2a, 3a)$. The supports are disjoint, while both vectors have norm one. Consequently

$$\|W_a - I\|_{H_G \to H_G} \geq \|(W_a - I) f_a\|_2 = \sqrt{2} \qquad (a > 0). \tag{17}$$

This lower bound suffices to exclude operator-norm continuity at zero; no optimal operator-norm value is needed.

6 Quartic and entropy obstructions

Theorem 6.1. *For every fixed $a \neq 0$, no bounded operator on $L^4(\gamma)$ can agree with W_a on all coherent vectors g_b, $b \in \mathbb{R}$. In particular, the L^2 unitary does not admit a bounded L^4 realization with its coherent action.*

Proof. Gaussian exponential moments give $\|g_b\|_4 = e^{b^2}$, while Theorem 5.1 gives $W_a g_b = g_{a+b}$. Thus

$$\frac{\|W_a g_b\|_4}{\|g_b\|_4} = e^{2ab+a^2}.$$

This diverges as $b \to +\infty$ if $a > 0$, or $b \to -\infty$ if $a < 0$. It contradicts any fixed bounded-operator norm. The argument uses only coherent functions, which belong to both L^2 and L^4. □

For a probability measure μ write

$$\mathrm{Ent}_\mu(F) = \int F \log F \, d\mu - \left(\int F \, d\mu\right) \log\left(\int F \, d\mu\right), \qquad 0 \log 0 = 0,$$

whenever the terms are finite. The roots in (9) have normalized squares, so their square entropies are relative entropies.

Theorem 6.2. *For $a = \sqrt{\lambda}(e^{\theta/2} - 1)$ one has*

$$\mathrm{Ent}_{p_\lambda}(r^2_{\lambda,\theta}) = \lambda(\theta e^\theta - e^\theta + 1), \tag{18}$$

$$\mathrm{Ent}_\gamma(g_a^2) = 2a^2, \qquad \|A_\lambda r_{\lambda,\theta}\|_2^2 = \|D g_a\|_2^2 = a^2. \tag{19}$$

For every $K \geq 0$ there is a positive θ such that

$$K \, \mathrm{Ent}_\gamma(g_a^2) < \mathrm{Ent}_{p_\lambda}(r^2_{\lambda,\theta}). \tag{20}$$

Hence exact root distances and derivative energies do not give a uniform upper comparison of these ordinary entropies.

Proof. The log of the Poisson likelihood is $\theta k - \lambda(e^\theta - 1)$. Its expectation under the tilted Poisson law uses $\mathbb{E}_{p_{\lambda e^\theta}} k = \lambda e^\theta$, giving (18). All absolute logarithmic moments are finite by Poisson exponential integrability. For the Gaussian root, $g_a^2 \gamma = N(2a, 1)$ and $\log g_a^2 = 2ax - 2a^2$, giving entropy $2a^2$. The identities $A_\lambda r = ar$ and $D g_a = a g_a$ give the energies because both roots have norm one.

Choose $\theta = 2K + 2$. Since $\theta > 0$,

$$\mathrm{Ent}_\gamma(g_a^2) = 2\lambda(e^{\theta/2} - 1)^2 \leq 2\lambda e^\theta,$$

whereas $\mathrm{Ent}_{p_\lambda}(r^2) = \lambda[(\theta - 1)e^\theta + 1] > 2K\lambda e^\theta$. This proves (20). □

Corollary 6.3. *For every $C \geq 0$ there is a real finite-support function f on $\mathbb{N}$ such that*

$$C\lambda \sum_{k \geq 0} p_\lambda(k)[f(k+1) - f(k)]^2 < \mathrm{Ent}_{p_\lambda}(f^2). \tag{21}$$

Such an f belongs to the original difference domain and to the original creation domain. Thus failure of a quadratic Poisson logarithmic Sobolev estimate already occurs on this common core.

Proof. By Theorem 6.2, choose $\theta > 0$ with $\mathrm{Ent}_{p_\lambda}(r^2) > Ca^2$, where $r = r_{\lambda,\theta}$. Set $f_M(k) = r(k)\mathbf{1}_{\{k<M\}}$ for $M \geq 1$. The L^2 tail tends to zero. The difference between Δf_M and Δr is zero below $M - 1$, equals $-r(M)$ at $M - 1$, and equals $-\Delta r(k)$ for $k \geq M$. Consequently

$$\|A_\lambda f_M - A_\lambda r\|_2^2 = \lambda p_\lambda(M-1) r(M)^2 + \lambda \sum_{k \geq M} p_\lambda(k)[\Delta r(k)]^2 \longrightarrow 0. \tag{22}$$

The second term is a tail of a convergent series. The first is a constant times $\lambda^M e^{\theta M}/(M-1)!$, which tends to zero by the ratio test. This checks the otherwise significant cutoff boundary in the actual maximal graph.

Since $r(k)^2|\log r(k)^2|$ is bounded by a constant times $e^{\theta k}(1+k)$, it is Poisson integrable. Therefore

$$\sum p_\lambda f_M^2 \log f_M^2 \longrightarrow \sum p_\lambda r^2 \log r^2, \qquad \sum p_\lambda f_M^2 \longrightarrow 1.$$

Continuity of $x \log x$ at 1 proves $\mathrm{Ent}(f_M^2) \to \mathrm{Ent}(r^2)$. Equation (22) gives $\|A_\lambda f_M\|_2^2 \to a^2$. The strict entropy-energy margin thus persists for all sufficiently large M, giving (21). Finally both difference and creation expressions of a finite-support sequence have finite support, so both original domain assertions hold directly. □

This obstruction concerns quadratic difference energy. It does not contradict the modified Poisson inequality with jump Bregman terms. For clarity, let $F = r^2_{\lambda,\theta}$. Its jump is $F(k+1) = e^\theta F(k)$, and hence

$$\lambda \sum_k p_\lambda(k) \left[F(k+1) \log \frac{F(k+1)}{F(k)} - F(k+1) + F(k) \right] = \lambda(\theta e^\theta - e^\theta + 1) = \mathrm{Ent}_{p_\lambda}(F).$$

The same exponentials therefore attain equality in the classical modified estimate [2], while excluding every fixed quadratic constant. Ordinary product structure determines which energy is relevant.

7 Positive quadratic curvature and the missing chain rule

The same original graphs identify the immigration–death expression

$$Lf(k) = \lambda[f(k+1) - f(k)] + k[f(k-1) - f(k)] = -A_\lambda^* A_\lambda f(k). \tag{23}$$

At zero the death term is zero. The last equality is pointwise whenever the two original graph memberships hold, and in particular on finite support. For arbitrary functions on $\mathbb{N}$ define, pointwise,

$$\Gamma(f,g) = \tfrac{1}{2}[L(fg) - fLg - gLf], \quad \Gamma(f) = \Gamma(f,f), \quad \Gamma_2(f) = \tfrac{1}{2}L\Gamma(f) - \Gamma(f, Lf).$$

Every expression uses only finitely many neighboring values. There is no summability assumption in a pointwise curvature statement. This quadratic jump calculus is different from diffusion chain calculus; Weber–Zacher [3, Example 5.13] already use the Poisson process to separate positive Bakry–Émery curvature from their entropy-adapted discrete curvature. We make no new curvature claim.

Theorem 7.1. *For every real function f on $\mathbb{N}$,*

$$\Gamma(f)(k) = \frac{\lambda}{2}[f(k+1) - f(k)]^2 + \frac{k}{2}[f(k) - f(k-1)]^2, \qquad \Gamma_2(f)(k) \geq \tfrac{1}{2}\Gamma(f)(k). \tag{24}$$

The constant $1/2$ is the largest constant valid uniformly in k and f, even if the tested functions are required to have finite support. Nevertheless, for every $C \geq 0$ there is a finite-support f with

$$C \int \Gamma(f)\, dp_\lambda < \mathrm{Ent}_{p_\lambda}(f^2). \tag{25}$$

Proof. Expanding $L(fg) - fLg - gLf$ gives

$$\Gamma(f,g)(k) = \tfrac{\lambda}{2}\Delta f(k)\Delta g(k) + \tfrac{k}{2}[f(k) - f(k-1)][g(k) - g(k-1)],$$

and hence the first formula. To see all terms in the second assertion, put $u = f(k+1) - f(k)$, $v = f(k) - f(k-1)$ and

$$A = f(k+2) - 2f(k+1) + f(k), \qquad B = f(k) - 2f(k-1) + f(k-2).$$

For $k \geq 2$, the neighboring first differences are $u + A$ and $v - B$, so

$$\begin{aligned} \Gamma(f)(k+1) &= \tfrac{\lambda}{2}(u+A)^2 + \tfrac{k+1}{2}u^2, \\ \Gamma(f)(k-1) &= \tfrac{\lambda}{2}v^2 + \tfrac{k-1}{2}(v-B)^2, \\ \Delta Lf(k) &= \lambda A - u - k(u-v), \\ Lf(k) - Lf(k-1) &= \lambda(u-v) - v - (k-1)B. \end{aligned}$$

Substitute these four identities into the definition of Γ_2. The linear cross terms in A, B cancel; the remaining u, v cross term is absorbed into $(u-v)^2$. The result is the exact decomposition

$$\begin{aligned} \Gamma_2(f)(k) = \frac{\lambda^2}{4}A^2 + \frac{k(k-1)}{4}B^2 + \frac{\lambda k}{2}(u-v)^2 \\ + \frac{3\lambda}{4}u^2 + \frac{k}{4}v^2. \end{aligned} \tag{26}$$

At $k = 0$ the terms with k vanish, and at $k = 1$ the B term vanishes; direct evaluation of the defining expressions gives the same formula at both endpoints. Negative-index values can be set to zero, since their terms have zero coefficient. Subtracting $\frac{1}{2}\Gamma(f)$ leaves the three nonnegative square terms and $\lambda u^2/2$, proving the bound.

For the count function $f(k) = k$, one obtains

$$\Gamma(f)(k) = \frac{\lambda + k}{2}, \qquad \Gamma_2(f)(k) = \frac{3\lambda + k}{4}, \qquad \frac{\Gamma_2(f)(k)}{\Gamma(f)(k)} \longrightarrow \frac{1}{2}.$$

Any proposed larger constant fails at some finite k. Replace this count function by a finite-support function agreeing with it on $\{0, \ldots, k+2\}$; both pointwise expressions at that k are unchanged. This proves the finite-support sharpness assertion as an elementary local deduction from the verified count witness.

Finally detailed balance $kp_\lambda(k) = \lambda p_\lambda(k-1)$ gives, for finite-support f,

$$\int \Gamma(f)\, dp_\lambda = \lambda \sum_{k \geq 0} p_\lambda(k)[\Delta f(k)]^2 = \|A_\lambda f\|_2^2.$$

Indeed the death-square sum reindexes to the birth-square sum, each contributing half of the last energy. Corollary 6.3 now gives (25), for the same finite-support domain on which the curvature estimate holds everywhere. □

The missing diffusion step can be seen at one atom. For the count function $\iota(k) = k$ and $\Phi(x) = e^x$,

$$\Gamma(\Phi \circ \iota)(0) = \frac{\lambda}{2}(e-1)^2 > \frac{\lambda}{2} = \Phi'(\iota(0))^2\Gamma(\iota)(0).$$

The strict inequality uses $e > 2$. Since $p_\lambda(0) > 0$, even an almost-everywhere version of that diffusion chain rule is false. The source energy $E_f(k) = \lambda[\Delta f(k)]^2$ is also different from $\Gamma(f)(k)$ pointwise: for $f = \iota$ at zero the values are λ and $\lambda/2$. Their integrals agree on the core, which does not make their pointwise identities interchangeable.

There is even a unit lower Bochner bound for this different forward energy. Direct evaluation gives $\Delta Lf = L\Delta f - \Delta f$. Applying $\frac{1}{2}L(h^2) = hLh + \Gamma(h)$ with $h = \Delta f$ therefore yields

$$\tfrac{1}{2}LE_f - \lambda\Delta f\,\Delta Lf = E_f + \lambda\Gamma(\Delta f) \geq E_f.$$

This identity and the sharp $1/2$ bound for the generator's Γ are both valid. Neither supplies the missing chain rule or the false quadratic entropy estimate. The original graph equivalence with Gaussian operators must be used together with the product obstruction in Proposition 2.2.

8 Full-domain recovery and a displaced vacuum

The preceding equivalence also identifies the exact range of creation. We give the scalar normalization explicitly, so that no weighted-field factor is hidden in the estimates. Write $p_n = \varphi_n^\lambda$ for the orthonormal Charlier basis of Section 2, with $p_0 = 1$, and set

$$C_\lambda f = \sum_{n\geq 0} \frac{\langle p_{n+1}, f\rangle}{\sqrt{n+1}} p_n, \qquad f^\circ = f - \mathbb{E}_\lambda f.$$

Proposition 8.1. *The bounded compact operator $C_\lambda : H_\lambda \to H_\lambda$ satisfies, for every $f \in H_\lambda$,*

$$C_\lambda f \in \mathrm{Dom}(A_\lambda^*), \quad A_\lambda^* C_\lambda f = f^\circ, \quad \|C_\lambda f\|_2^2 = \sum_{q\geq 1} \frac{|\langle p_q, f\rangle|^2}{q} \leq \|f^\circ\|_2^2.$$

Creation is injective and has range $1^\perp$. Furthermore

$$\|f^\circ\|_2 \leq \|A_\lambda f\|_2, \qquad f \in \mathrm{Dom}(A_\lambda),$$

with equality for $f = p_1$.

Proof. Parseval proves convergence of the displayed series and its norm identity. The finite sums belong to the polynomial graph core; their creation images converge to f°. Closedness of the original creation graph therefore proves the claimed full-domain identity. If $u = \sum u_n p_n$ belongs to that graph, its creation coefficients are $\sqrt{n+1}u_n$ in mode $n+1$. Thus $A_\lambda^* u = 0$ forces every $u_n = 0$, and every creation image has zero constant coefficient. The recovery identity proves the converse range inclusion and uniqueness of the preimage. The maximal annihilation graph gives

$$\|A_\lambda f\|_2^2 = \sum_{q\geq 1} q|\langle p_q, f\rangle|^2 \geq \sum_{q\geq 1} |\langle p_q, f\rangle|^2 = \|f^\circ\|_2^2.$$

Only the first mode is needed for equality. Finally truncate recovery to input modes $1, \ldots, N$. This has finite rank, and the omitted operator has norm at most $(N+1)^{-1/2}$ by the same coefficient identity. Thus C_λ is an operator-norm limit of finite-rank operators and is compact. In fact the bound is attained on p_{N+1}. □

Now fix a physical intensity change $\mu = \lambda e^\theta$, let $V = V_{\lambda,\theta} : H_\mu \to H_\lambda$ be the root multiplication unitary, and put $a = \sqrt{\mu} - \sqrt{\lambda}$ and $r = V1$. The vector r has norm one, but its scalar mean need not be one. Define

$$P_r f = f - \langle r, f\rangle r, \qquad C_r = VC_\mu V^{-1}.$$

Corollary 8.2. *For every $f \in H_\lambda$,*

$$(A_\lambda^* - a)C_r f = P_r f, \qquad \|C_r f\|_2 \le \|P_r f\|_2.$$

The shifted creation graph has range $r^\perp$ and zero kernel; $C_r f$ is its unique preimage of $P_r f$. The shifted annihilation graph has kernel $\mathbb{R}r$ and satisfies

$$\|P_r f\|_2 \le \|(A_\lambda - a)f\|_2, \qquad f \in \mathrm{Dom}(A_\lambda).$$

The operator C_r is compact, with the same finite-rank tail bounds.

Proof. The full-domain gauge equalities proved earlier give $(A_\lambda - a)V = VA_\mu$ and $(A_\lambda^* - a)V = VA_\mu^*$. The orthogonal projection onto constants in H_μ becomes projection onto $\mathbb{R}r$ after conjugation. Conjugate every assertion of Proposition 8.1 by V. For the annihilation kernel, its coefficient energy is zero exactly when all positive modes vanish. Conjugating the finite-rank approximants proves compactness and the tail bounds. Addition of the bounded scalar operator $-aI$ leaves the respective maximal operator domains unchanged. □

Under U_λ, the vacuum r becomes $g_a(x) = e^{ax-a^2}$. Consequently the Gaussian operator $\delta - a$ has exact range $g_a^\perp$, its inverse on that range is compact, and $\ker(D - a) = \mathbb{R}g_a$. These are one-coordinate statements, obtained by the same unitary conjugation. They do not assert injectivity of divergence on a general Gaussian vector-field space. At fixed λ, physical tilts give $a > -\sqrt{\lambda}$; varying λ realizes every real Gaussian displacement. The vacuum projection, not subtraction of the ordinary scalar mean, is the displaced centering operation.

9 Two intensity regimes beyond the Hilbert norm

The preceding sharp Hilbert estimates do not yield a lower bound for the ordinary L^p norm at other exponents. This can be checked on a single actual compensated count, without a process construction. Let N_r have Poisson law of intensity $r > 0$, and put

$$Z_r = N_r - r, \quad b(r) = \|Z_r\|_{4/3}, \quad s(r) = \min\{r^{1/2}, r^{3/4}\}.$$

Proposition 9.1. *For every $r > 0$,*

$$\tfrac{1}{4}\min\{r^2, r^3\} \le b(r)^4 \le 4\min\{r^2, r^3\}, \qquad \frac{s(r)}{\sqrt{2}} \le b(r) \le \sqrt{2}\, s(r).$$

No positive constant c satisfies

$$c\|1\|_{L^{4/3}(rp_r)} \le \|Z_r\|_{L^{4/3}(p_r)} \quad \textit{for every } r > 0.$$

Here Z_r is precisely the original, unnormalized Poisson divergence of the constant field 1.

Proof. The generating function $\mathbb{E}t^{N_r} = e^{r(t-1)}$ gives factorial moments $\mathbb{E}(N_r)_j = r^j$. Expanding ordinary powers in falling factorials gives

$$\mathbb{E}Z_r^2 = r, \qquad \mathbb{E}Z_r^4 = 3r^2 + r.$$

All moments are finite. Hölder with exponents 4/3 and 4, applied to $|Z_r|\,|Z_r|$, yields

$$r \le b(r)(3r^2 + r)^{1/4}, \qquad b(r)^4 \ge \frac{r^3}{3r+1}.$$

For $r \le 1$ the denominator is at most 4; for $r \ge 1$ it is at most $4r$. This proves the lower estimate. Since the law is a probability, monotonicity of norms gives $b(r) \le \sqrt{r}$. For the other upper bound, Hölder applied to $|Z_r|^{2/3}|Z_r|^{2/3}$ with exponents $3/2$ and 3 gives

$$\|Z_r\|_{4/3} \le \|Z_r\|_1^{1/2} \|Z_r\|_2^{1/2}.$$

The elementary inequality $\mathbb{E}|N_r - r| \le 2r$ then gives $b(r) \le \sqrt{2}\, r^{3/4}$. Combining both upper bounds proves the assertion; taking fourth roots gives the stated comparison with $s(r)$.

The original field space has control measure rp_r and its unnormalized divergence is $\delta_r u(k) = ku(k-1) - ru(k)$. For $u = 1$ this equals Z_r. This field belongs to the original maximal L^2 divergence graph: indeed the scalar unitary field normalization maps it to $\sqrt{r}\, p_0$, and $A_r^*(\sqrt{r}\, p_0) = \sqrt{r}\, p_1 = Z_r$. On the other hand $\|1\|_{L^{4/3}(rp_r)} = r^{3/4}$. Choose $r = n^4$ with integers $n \ge 1$. The sharper probability-norm upper bound and the lower estimate give

$$\frac{1}{\sqrt{2}\, n} \le \frac{\|\delta_{n^4} 1\|_{4/3}}{\|1\|_{L^{4/3}(n^4 p_{n^4})}} \le \frac{1}{n}.$$

The ratio tends to zero, excluding every proposed $c > 0$. □

The two scales have different meanings: $r^{3/4}$ describes rare counts for $r \le 1$, while $r^{1/2}$ describes the large-intensity regime. Only the latter regime is needed for the displayed failure of a plain $L^{4/3}$ lower bound. There is no contradiction with the L^2 estimate, where the same constant input has norm $\sqrt{r}$. Every estimate transfers to a count on another probability space having this Poisson law, since all expressions depend only on that one law; no independence of several counts is required.

This elementary diagnostic belongs to established Poisson integration theory. Dirksen's Theorem 1.1 [4] gives much stronger maximal stochastic integral estimates with sums and intersections of square-function and L^p-type spaces. For a scalar deterministic indicator of control mass r and exponent $4/3$, its component scales are $\sqrt{r}$ and $r^{3/4}$; the sum-space norm selects their minimum up to constants. The proposition proves the count comparison directly. It does not construct a general random measure or claim that a single plain $L^{4/3}$ norm is the appropriate integrand space.

10 Exact provenance and mathematical scope

All formal source references below use the frozen verified commit

`bab688bb6140cd6316c80680095a914ea0d346db`

Its original audit receipt and unchanged source hashes accompany the paper in the baseline-verification JSON file. The publication editor has not rerun a research build or altered the verified library. The accompanying `theorem-map.json` records exact declarations, line numbers and public baseline links. The source groups cover three connected layers:

- Complete Charlier coordinates, weighted scalar geometry and exact Poisson spectral graphs; the Gaussian Hermite adjoint graphs.
- Half-density source gauges, Gaussian displacement from Poisson, coherent identification, group law, continuity and physical action.

- Tilt Hellinger geometry, operator-norm discontinuity, quartic unboundedness, coherent entropy distortion, finite-core failure and original Poisson jump curvature.

The recovery and shifted Poincaré results are scalar normalizations of the original weighted-field graphs. The two-regime count estimates use actual Poisson laws. Compactness and the Gaussian range statements above are also proved directly by coefficients and unitary conjugation.

The finite-core entropy proof specializes the verified cutoff argument to an explicit coherent root. Generating functions, weak-distributional domains and the dense coherent-span identification reorganize the exact formal graph results into human proofs; every displayed line is not a separate formal declaration. An accompanying standard-library Python verifier checks the Gamma-two decomposition by exact rational polynomial arithmetic, including both boundary cases; it is a transcription check. The positivity witness is an elementary analytic deduction, distinguished in the theorem map.

The setting is one Poisson count and one Gaussian coordinate. The results do not transfer ordinary nonlinear equations, a Markov order or general entropy comparisons. This is a classical formalization companion; novelty certification and external submission remain separate questions.

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